

VEDANT PAWAR

PYTHON DEVELOPER & AI/ML ENGINEER

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PROFESSIONAL SUMMARY

Results-driven Python Developer and AI/ML Engineer with hands-on experience building production-grade NLP systems, LLM-integrated backends, and real-time Computer Vision applications (YOLO, OpenCV). Architected the backend for a 7-domain autonomous AI robot using FastAPI, LangChain, RAG pipelines, and Docker, driving 25% workflow efficiency gains. Proficient in cloud deployment, REST APIs, and scalable system design, actively seeking impactful Python/AI backend or ML Engineer roles.

CORE TECHNICAL SKILLS

Languages & Backends:	Python, FastAPI, Flask, Django, JavaScript, REST APIs, SQL (MySQL, PostgreSQL), Async Programming
AI / ML / NLP:	LLM Integration, LangChain, RAG Pipelines, Prompt Engineering, Vector Databases (ChromaDB), Hugging Face, OpenCV, YOLO, SSD, Faster R-CNN, NLP, Sentiment Analysis
Voice & UI:	gTTS, ElevenLabs API, Gradio, Streamlit, React (Basics), HTML5, CSS3
DevOps & Cloud:	Docker, Docker Compose, AWS (EC2, S3 deployment patterns), Git, GitHub, CI/CD Pipelines (GitHub Actions)
Core CS Concepts:	Data Structures & Algorithms (DSA), OOP, Microservices, Scalable Architecture, API Design Patterns, Backend Optimization

PROFESSIONAL EXPERIENCE

AI/ML & IoT Intern – Incubators (CREVA AI Robot Project)	Jan 2026 – Mar 2026
• Pipeline Architecture: Designed NLP-based interaction pipelines for the CREVA AI Robot across 7 functional domains, enabling real-time AI request processing and improving overall workflow efficiency by 25% (reducing average task completion time from ~40s to ~30s) utilizing Computer Vision techniques with OpenCV and PIL/Pillow. • API Optimization: Integrated 5+ AI modules with web interfaces via RESTful APIs, reducing inter-service API latency by ~30% through payload optimization and connection pooling. • Scalable Deployment: Designed and implemented a scalable AI+IoT backend architecture supporting modular, hot-swappable deployment workflows, enabling 3 new AI modules to be added without system downtime. • Containerization: Containerized the FastAPI backend using Docker, standardizing local development and enabling consistent, reproducible deployment environments across a 5-person engineering team. • Agile Execution: Collaborated in an Agile/Scrum environment across 6 two-week sprints, maintaining 100% on-time sprint delivery while iterating on real-time AI system integration.	

KEY PROJECTS

CREVA AI System – Multi-Domain Autonomous AI Robot
• Multi-Domain Orchestration: Deployed NLP-driven interaction pipelines enabling real-time, context-aware AI communication across 7 independent functional domains, serving 20+ concurrent request types and incorporating Computer Vision with YOLO, SSD, and Faster R-CNN. • LLM Integration: Integrated multiple LLM and AI APIs into a unified, scalable FastAPI backend with LangChain orchestration, supporting hot-swappable modular AI workflows and reducing integration effort by ~40%. • System Reliability: Engineered modular backend architecture with clean separation of concerns containerized with Docker, implementing robust API communication with error handling, retry logic, and async request queuing to achieve 99%+ uptime during active live testing.

EDUCATION

Bachelor of Computer Applications (BCA)	2022 – 2025
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CERTIFICATIONS & TRAINING

- **Python for Everybody** – Coursera
- **FastAPI & REST API Development** – Self-directed (3 production projects completed)
- **LangChain for LLM Application Development** – DeepLearning.AI (*In Progress*)
- **AWS Cloud Practitioner Essentials** – (*In Progress; EC2 & S3 deployment patterns applied in projects*)
- **Docker & Containerization** – Applied across 3 production GitHub repositories